CSCI 5526

Classification Problems in Healthcare

Announcements

- MP1 due on 9/25 at 11.59pm!
- Final project survey: Grid posted!

Student Questions

Deidentification

Level	Description	Privacy Strength	Data Utility	Notes
Raw Data	Identifiable data with direct identifiers intact	❌ None	✅ High	Fully exposed
Pseudonymization	Replace identifiers with codes or keys	🔶 Low	✅ High	Re-identifiable if key exists
De-identification	Remove direct identifiers; quasi-identifiers remain	🔶 Low–Medium	✅ High	Can be linked with external data
K-Anonymity	Each record is indistinguishable from ≥k others	⚖️ Medium	⚖️ Medium	Vulnerable to background knowledge
L-Diversity	Groups must have diverse sensitive attributes	⚖️ Medium–High	⚖️ Medium	Improves on k-anonymity
T-Closeness	Group distribution of sensitive attributes ≈ overall distribution	✅ High	🔶 Lower	Stronger, but reduces detail
Full Anonymization	All identifiers and quasi-identifiers removed; often aggregated	✅ Very High	❌ Low	Irreversible, but limited research use
Differential Privacy	Add mathematical noise to results/queries	✅ Very High (provable)	⚖️ Medium	Balances privacy with statistical utility

Student Questions

In practice, do doctors also consider financial factors? Would these factors influence how doctors make judgments, positively or negatively?

Positives:

1. Over 60% of physicians report routinely weighing affordability when prescribing drugs or ordering tests.
2. Financial models that reward quality and efficiency over sheer volume incentivize doctors to focus on preventive care
3. Awareness of macro-level healthcare costs forces doctors to treat scarce resources (like ICU beds or expensive MRI scans) with greater care.

Negatives:

1. In traditional fee-for-service payment systems, doctors or hospital groups earn more money when they do more procedures.
2. Conversely, in fixed-budget systems (like certain HMOs or capitated insurance plans), a clinic or hospital loses money if they order too many expensive services.
3. Doctors are often constrained by what insurance policies will approve.

Student Questions

Do you think the learning healthcare system framework sometimes takes too much time per cycle to adapt to new discoveries?

Yes.

1. The "Performance-to-Data" Bottleneck: Aggregating clean, normalized data across fragmented Electronic Health Record (EHR) systems often takes months or years.
2. The "Data-to-Knowledge" Bottleneck: Translating raw clinical data into actionable, generalized science requires deep statistical analysis, peer validation, and rigorous ethics reviews.
3. The "Knowledge-to-Performance" Bottleneck: Convincing busy clinicians to change behavioral habits, altering software decision-support tools, and shifting organizational culture requires massive effort.

Student Questions

When do we know that our model is robust for the particular type of data (single hospital/multiple hospitals)?

Single hospital: internal validation stability

Multiple hospitals: external validation stability

What are the best ways to clean the data (like the clinical data that we were talking about in class)? Good packages/rules/methods?

Depends on the use-case.

pandas & numpy – drop_duplicates(), fillna()

janitor – for standardizing names

scikit-learn – imputation

Student Questions

We talked about data a lot today, and I heard from my friend who works in a health insurance company that they are allowed to put PII/personal health information in their LLM. Is this compliant with regulation, if so, how? Do they just have to use "incognito mode" if using a 3rd party service? or do it on model that is run on prem?

Health insurance companies handle PHI using two primary architectural paths

1. Enterprise Third-Party Cloud Models - if—and only if—the provider signs a Business Associate Agreement (BAA) – a legally binding contract where the AI vendor agrees to safeguard the health data under HIPAA standards.
2. On-Premises or Private Cloud Deployment

Student Questions

Are there any other cases where incorrect assumptions on population have caused problems. For example, with the case of heart rates including both adult and newborn populations.

-- For decades, in pharmaceutical trials, children were treated as "small adults." Pediatric dosing guidelines were frequently calculated using a simple fraction of the adult dose based strictly on body weight or surface area.

If a model is trained and gives a better outcome and clinicians use it in their treatment and the new improved data is collected and used to retrain the model how can we prevent bias?

-- Do frequent checks to actively interrupt and audit every stage of the data lifecycle.

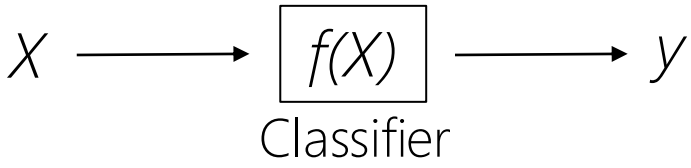
Audit new data, apply algorithmic safeguards, maintain continuous governance

Outline

- What is classification?
- Examples in healthcare
- Ingredients of a classification model
- Training, validation, testing
- Generalization
- Evaluation metrics
- Imbalanced data
- Deep learning vs classical ML, how to decide?

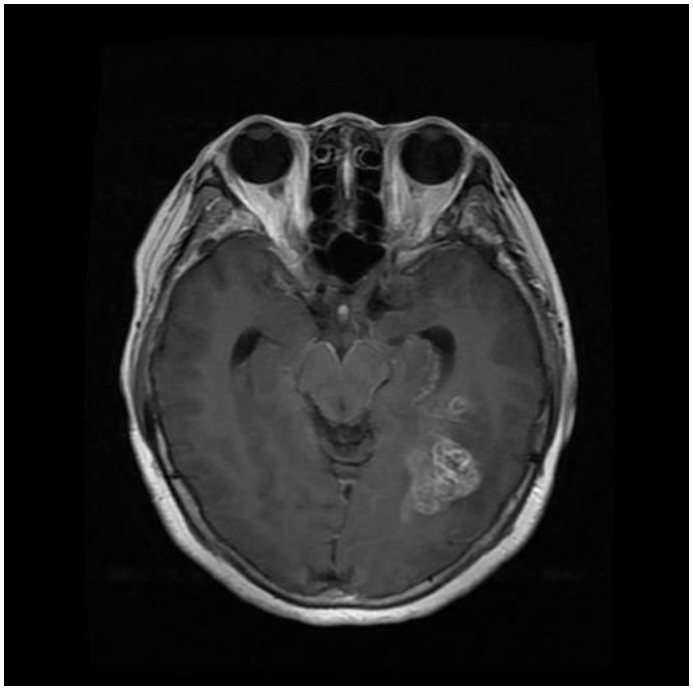
What is Classification?

- Given an input *X*, assign it to class *y*.
  - *X* can be a vector, image, text, etc.
  - All possible classes (or values) of y should be known ahead of time.
  - Usually supervised: i.e., we have example pairs $((x_1, y_1), \ldots, (x_N, y_N))$



- Binary classification: *y* is one of two values ('yes' or 'no')
- Multiclass classification: *y* is one of multiple values
- Regression is an extension: *y* is in a continuous range

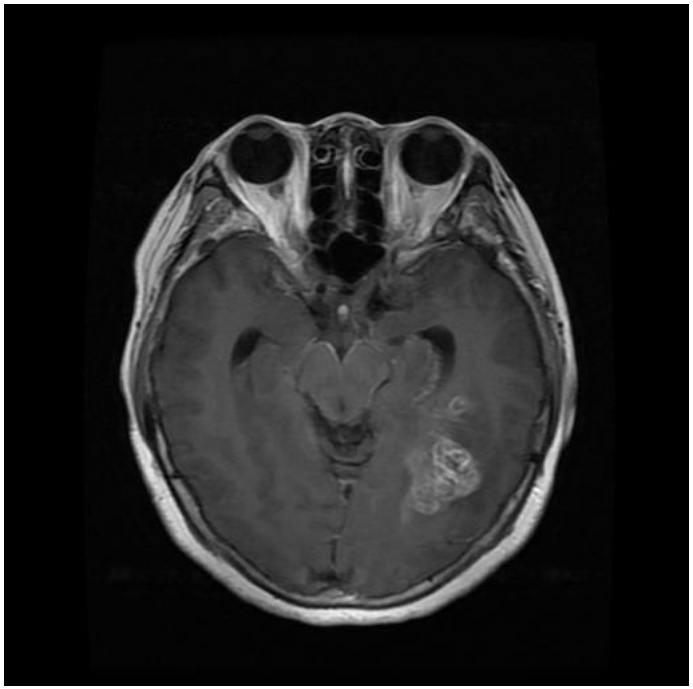
Example 1: Tumor Classification



Brain MRI image

⟶ Tumor present or not

Example 2: Tumor Segmentation



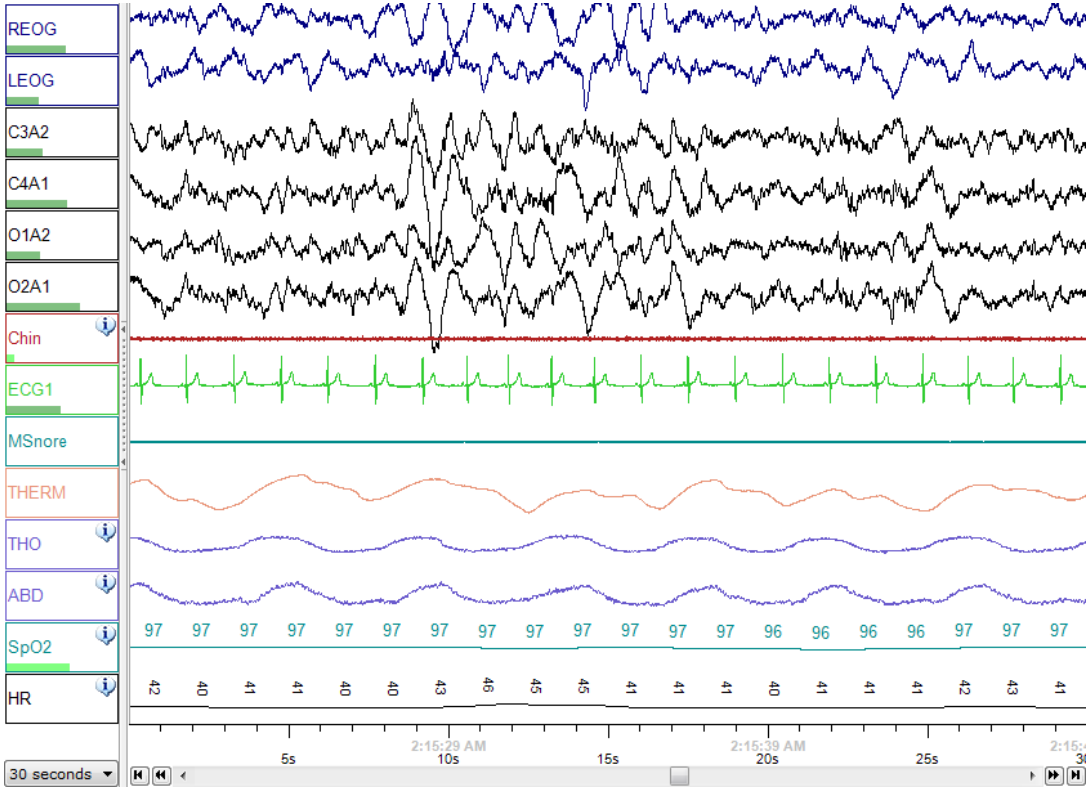
Brain MRI image



Brain MRI image

We classify each pixel!

Example 3: Sleep Stage Classification



Polysomnography data

Awake

N1

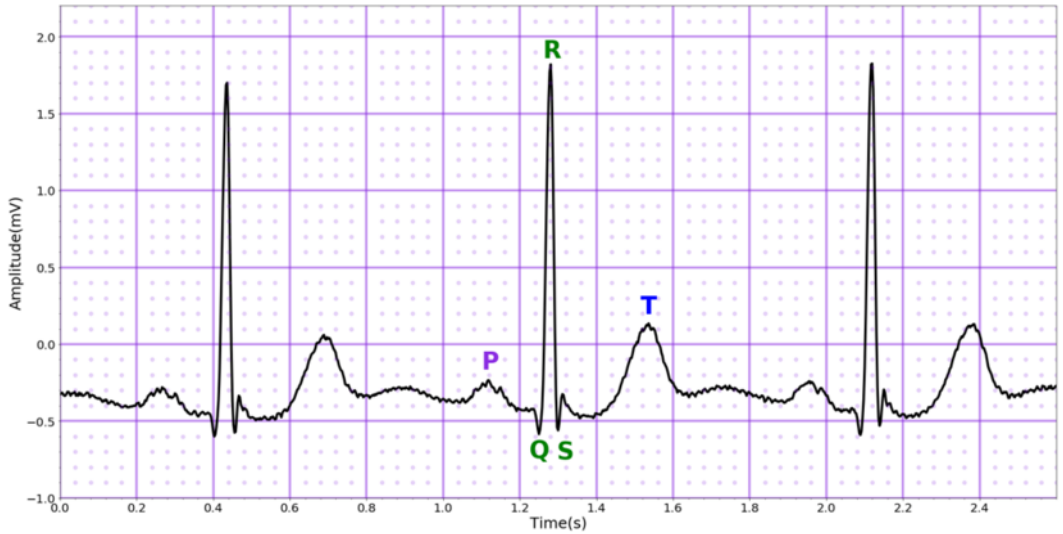
N2

N3

REM

Multi-class Classification!

Example 4: LVEF Estimation



→ Cardiac output

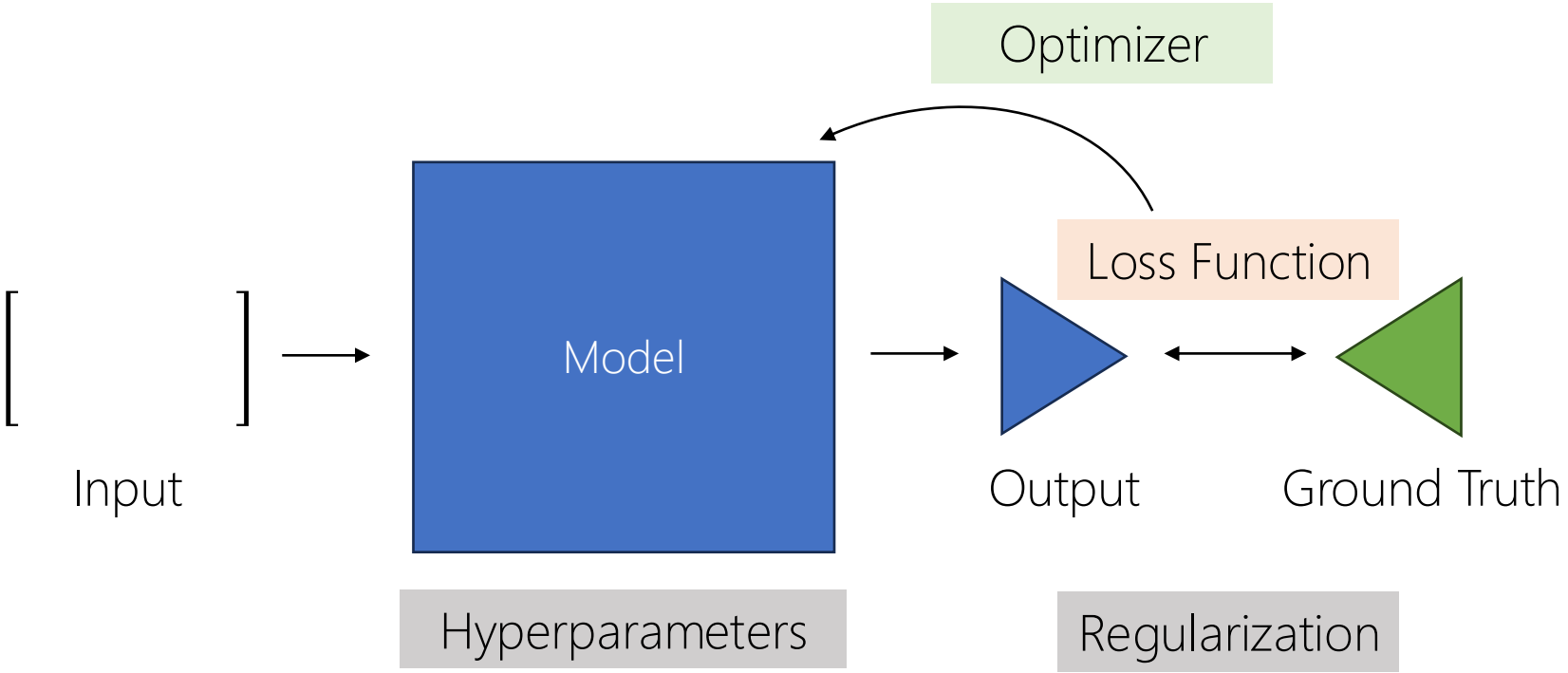
Regression!

Example 5: Treatment Recommendation

Status of patient:
Julie is worse today.
Target Symptoms:
Julie reports that depressive symptoms continue. Her symptoms, she reports, are more frequent or more intense. Anergia is present. Increased symptoms of anhedonia are present. Julie's difficulty with concentrating has not changed. Julie reports that she continues to feel sad. Guilty feelings are described by Julie. 'I should have been with my sister, I had no idea she was suicidal.' Sleep has improved with the use of PRN Ambien CR at HS. Julie convincingly denies suicidal ideas or intentions.
Basic Behaviors:
Medication has been taken regularly. She needs help with ADLs. When she attends activities participation is minimal. Prn's are used occasionally and are described as effective for her headaches. Impulsive behaviors are occurring, but less frequently. Julie has diminished food and fluid intake. Julie has not been confused. A good night's sleep is described.
Additional Signs or Possible Side Effects:
Sedative effects of the medication are described. Patient reports a dry mouth. No other side effects are reported or in evidence.
MENTAL STATUS:
Julie presents as glum, downcast, inattentive, minimally communicative, and looks unhappy. She appears listless and anergic. She appears downcast. Thought content is depressed. Slowness of physical movement helps reveal depressed mood. Facial expression and general demeanor reveal depressed mood. She denies having suicidal ideas. There are no apparent signs of hallucinations, delusions, bizarre behaviors, or other indicators of psychotic process. Associations are intact, thinking is logical, and thought content appears appropriate. There are signs of anxiety. Patient is fidgety in a way that is suggestive of anxiety.
Special Circumstances:
Julie continues to have an unsteady gait, especially after midnight. Call light is within her reach. She has been instructed to ring for the nurse to assist her when ambulating to bathroom.
Vital Signs:
Sitting blood pressure is 150
/ 85. Sitting pulse rate is 80. Respiratory rate is 18 per minute. Temp. is 98+ F. Weight is 155 lbs. (70.3 Kg).

→ Hospitalize?

Basic Ingredients of a Classification Model



Input Data

- Raw data
  - E.g., images, signals
- Tabular data (processed features)
  - E.g., image features, signal features, gene expression
- Sequence data
  - E.g., timeseries, text, protein sequences
- Spatial data
  - E.g., graphs, maps

Models

- Raw data (signals, images)
  - Convolutional neural networks
- Tabular data (processed features)
  - Linear classifiers: SVM (vanilla), linear/logistic regression
  - Non-linear classifiers: nearest neighbor, SVM(kernel), neural networks
  - Tree-based classifiers: decision trees, random forests
- Sequence data
  - E.g., graphical models, sequence models (RNN, LSTM, attention)
- Spatial data
  - E.g., graphical models, graph neural networks

Outputs and Loss Functions

Outputs

- Classification
  - Categorical ({0,1} or {0,1,…,K})
  - Probabilistic ([0,1])
- Regression
  - Real values

Loss Functions

- Binary classification
  - Logistic loss (or binary cross entropy)
  - Hinge loss
  - Huber loss
- Multiclass classification
  - Cross entropy
  - Convert to binary (one vs all)
- Regression
  - Mean square error
  - Mean absolute error
  - Huber loss

Optimizers

- Gradient descent, accelerated gradient descent (momentum)
- Second order methods (Newton)
- Stochastic gradient descent (or batch)
  - ADAM, AdaGrad, RMSProp

Regularization

- Used to add constraints to the optimization task.
  - Avoid overfitting
  - Mechanistic/physics constraints

- Examples:
  - L1
  - L2
  - Elastic net
  - Dropout
  - Ordinary differential equations

Hyperparameters: Examples

- Every classifier has some hyperparameters.
- Logistic regression
  - Penalty function: ['none', 'l1', 'l2', 'elasticnet']
- SVM
  - Kernels: ['linear', 'poly', 'rbf', 'sigmoid']
  - Penalty C: [100, 10, 1.0, 0.1, 0.001]
- Neural networks
  - # layers
  - # neurons in each layer

Generalization



Training set (labels known)



Test set (labels unknown)

- How well does a learned model generalize from the data it was trained on to a new test set?

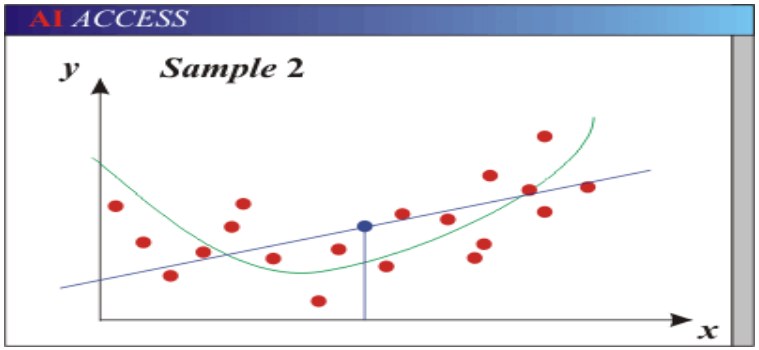
Slide credit: L. Lazebnik

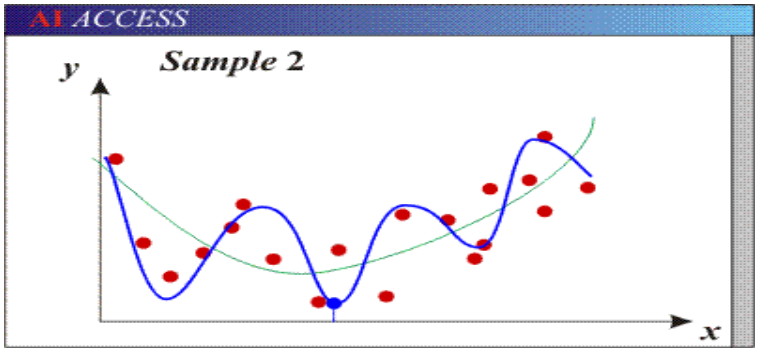
Generalization

- Components of generalization error
  - **Bias:** how much the model differs from the true model?
    - Error due to inaccurate assumptions/simplifications made by the model
  - **Variance:** how much models estimated from different training sets differ from each other
- **Underfitting:** model is too "simple" to represent all the relevant class characteristics
  - High bias and low variance
  - High training error and high test error
- **Overfitting:** model is too "complex" and fits irrelevant characteristics (noise) in the data
  - Low bias and high variance
  - Low training error and high test error

Slide credit: L. Lazebnik

Bias-Variance Trade-off

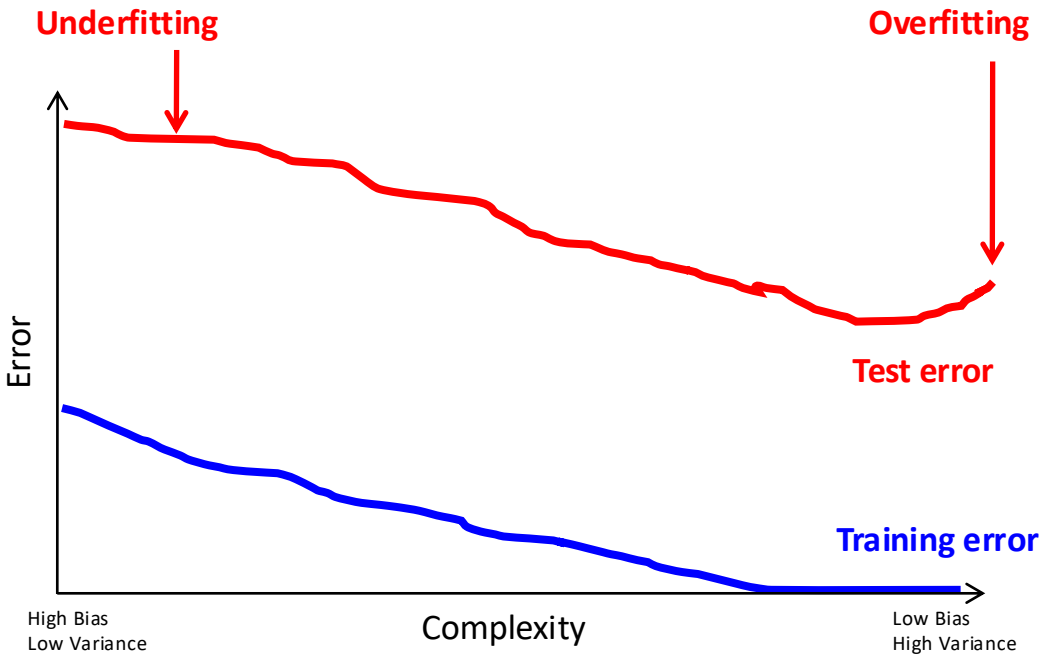




- Models with too few parameters are inaccurate because of a large bias (not enough flexibility).
- Models with too many parameters are inaccurate because of a large variance (too much sensitivity to each sample).

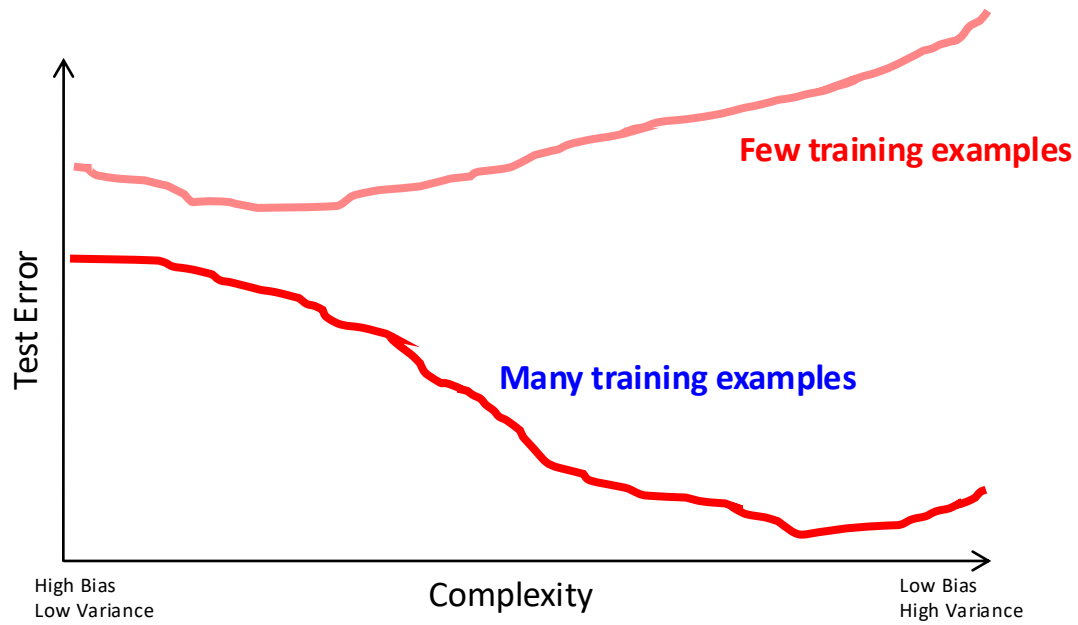
Slide credit: D. Hoiem

Bias-variance tradeoff



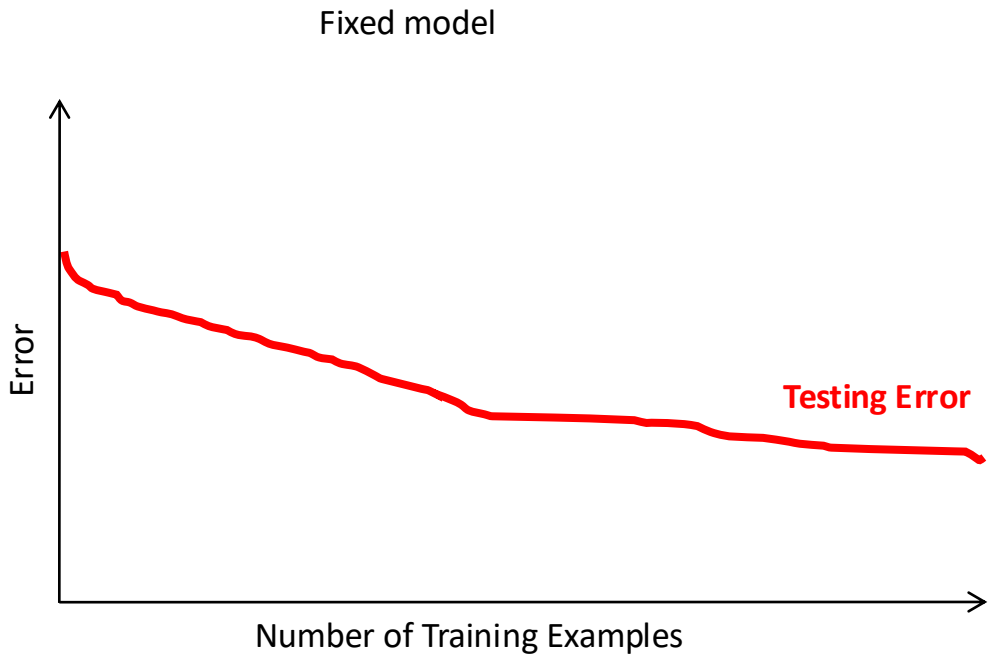
Slide credit: D. Hoiem

Bias-variance tradeoff



Slide credit: D. Hoiem

Effect of Training Size



Slide credit: D. Hoiem

Remember…

- No classifier is inherently better than any other.
  - You need to make assumptions to generalize.

- Three kinds of error
  - Inherent: unavoidable
  - Bias: due to over-simplifications
  - Variance: due to inability to perfectly estimate parameters from limited data

Slide credit: D. Hoiem

How to reduce variance?

- Choose a simpler classifier
- Cross-validate the parameters
- Get more training data

Slide credit: D. Hoiem

Cross-validation



Evaluation Metrics

- Classification
  - Confusion matrix
  - Sensitivity, specificity
  - Precision, recall
  - Accuracy
  - Area under ROC curve
- Regression
  - Mean absolute error
  - Mean square error
  - R-squared

Definitions

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$

$$ErrorRate = 1 - accuracy$$

$$Precision = Positive\ Predictive\ Value = \frac{TP}{TP + FP}$$

$$Recall = Sensitivity = TP\ Rate = \frac{TP}{TP + FN}$$

$$Specificity = TN\ Rate = \frac{TN}{TN + FP}$$

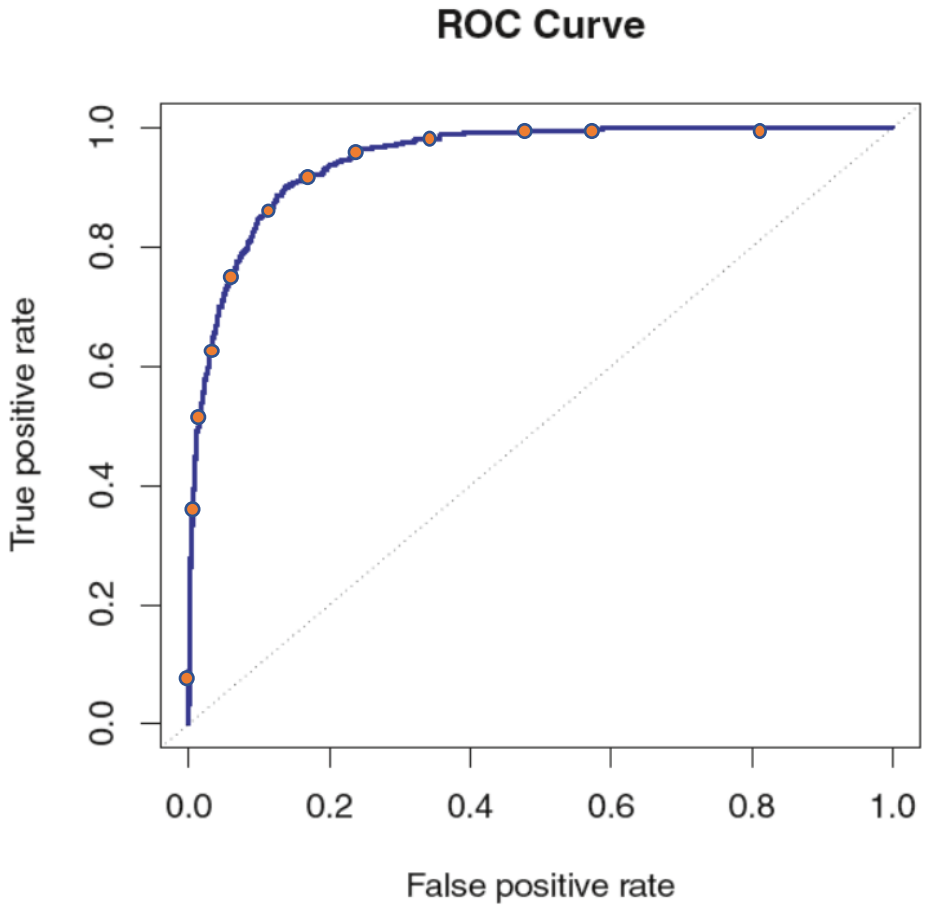
$$FP\ Rate = \alpha = \frac{FP}{TN + FP} = 1 - specificity$$

$$FN\ Rate = \beta = \frac{FN}{FN + TP} = 1 - sensitivity$$

$$Power = sensitivity = 1 - \beta$$

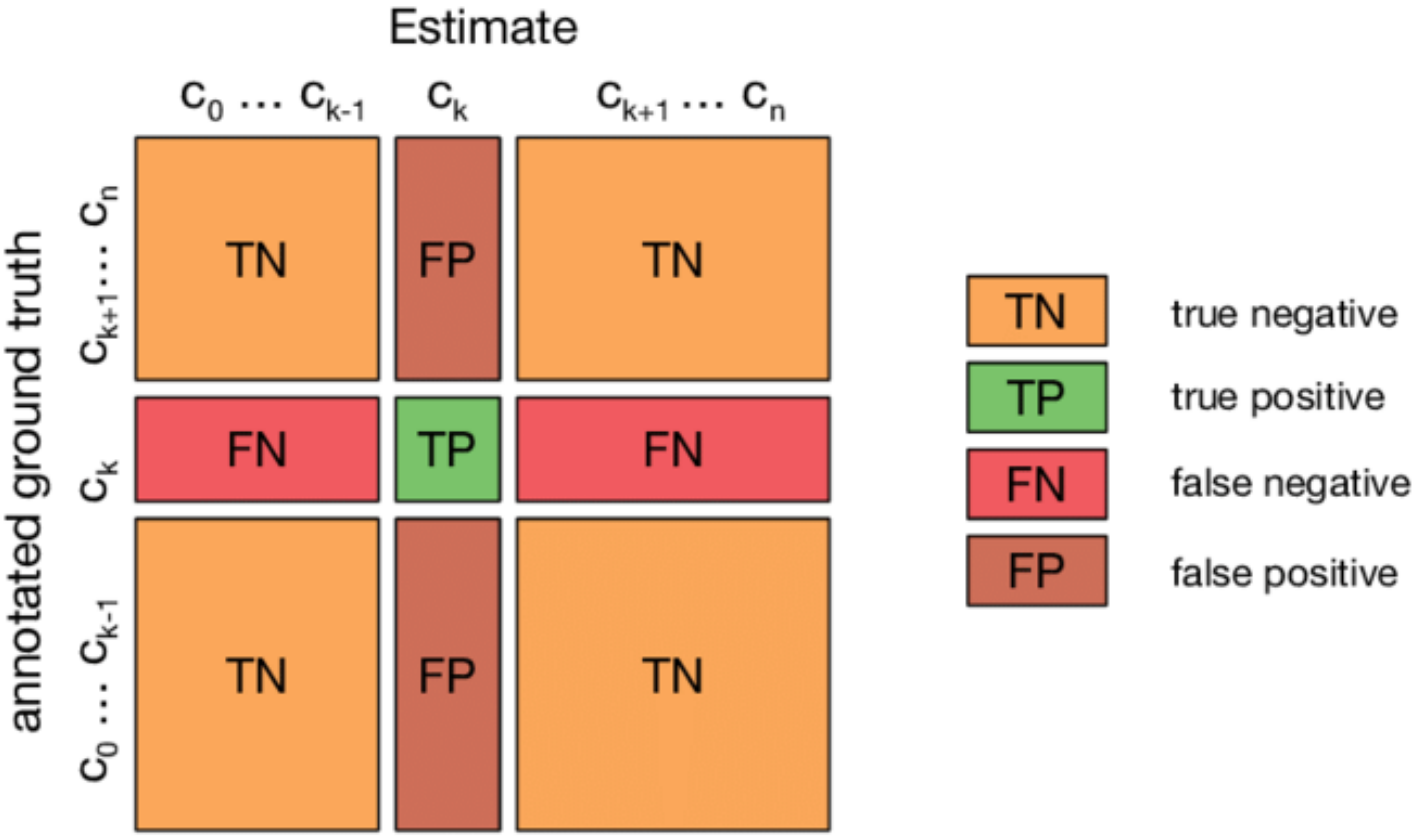
$$\text{F - measure (F)} = \frac{2rp}{r + p} = \frac{2a}{2a + b + c}$$

ROC Curve (probabilistic outputs)



- Only for binary classification with probabilistic output.
- Different TPR, FPR are calculated by varying the classification threshold.
- Area under ROC curve is a robust measure of performance (aka AUC or AUROC).

Multi-class Setting



Macro vs Micro Metrics

- Macro – treats each class with the same weight
- Micro – treats each sample with the same weight

To illustrate why, take for example precision $Pr = \frac{TP}{(TP+FP)}$. Let's imagine you have a *One-vs-All* (there is only one correct class output per example) multi-class classification system with four classes and the following numbers when tested:

- Class A: 1 TP and 1 FP
- Class B: 10 TP and 90 FP
- Class C: 1 TP and 1 FP
- Class D: 1 TP and 1 FP

You can see easily that $Pr_A = Pr_C = Pr_D = 0.5$, whereas $Pr_B = 0.1$.

- A macro-average will then compute: $Pr = \frac{0.5+0.1+0.5+0.5}{4} = 0.4$
- A micro-average will compute: $Pr = \frac{1+10+1+1}{2+100+2+2} = 0.123$

Dealing with Class Imbalance

- What is class imbalance?
  - Drastic variability in the prevalence of classes
  - Very common in healthcare

- Consider the case of screening for lung cancer:
  - Sample: individuals over the age of 30
  - Prevalence: 1 in 16
  - Data: chest X-ray
  - 1000 images will include approximately 60 positive cases

What is the problem?

Classifier A

		PREDICTED	
		Class=Yes	Class=No
ACTUAL	Class=Yes	0	10
	Class=No	0	990

Accuracy: 99%

Classifier B

		PREDICTED	
		Class=Yes	Class=No
ACTUAL	Class=Yes	10	0
	Class=No	200	790

Accuracy: 80%

Solution: Try other Metrics

Classifier A

		PREDICTED	
		Class=Yes	Class=No
ACTUAL	Class=Yes	0	10
	Class=No	0	990

Accuracy: 99% Sensitivity: 0% Specificity: 100%

Classifier B

		PREDICTED	
		Class=Yes	Class=No
ACTUAL	Class=Yes	10	0
	Class=No	200	790

Accuracy: 80% Sensitivity : 100% Specificity : ~80%

General Guidelines

- Beware of class imbalance.
- Understand the impact of each type of error.
- Try multiple metrics.
- Work with domain experts to optimize performance or reformulate the question.

When to use deep learning?

- Deep vs shallow learning
  - Shallow learning: learning from extracted features
  - Deep learning: learning from raw data
- Sample size:
  - >few thousands → deep learning
- Data type:
  - Raw data → deep learning

Exit pass

https://z.umn.edu/ml4h_exitpass

